

# MOHAMMED ALI

Computer Science Engineering (AI & ML) Student

## SELECTED PROJECTS

### BuildFin— Construction Financial Intelligence Platform

**Tech:** Next.js · React · TypeScript · Prisma ORM · PostgreSQL · NextAuth.js · Tailwind CSS

- Developed a construction financial intelligence platform that correlates project progress with budgets, BOQs, expenses, and material usage to generate actionable financial insights and decision support.
- Designed modular financial intelligence engines for adaptive financial baselines, burn-rate monitoring, benchmarking, and risk detection while maintaining a clear separation between business logic and the user interface.
- Built a deterministic Excel import pipeline featuring concept-based column mapping, transactional validation, preview workflows, and audit logging for reliable financial data ingestion.
- Implemented secure backend infrastructure using Next.js Server Actions, NextAuth.js, OTP verification, session management, rate limiting, and account lockout mechanisms to strengthen application security.

### MicroTrace— Intelligent Microplastic Analysis System

**Tech:** React · TypeScript · Vite · FastAPI · Python · YOLOv8 · OpenCV · PyTorch · NumPy

- Developed an intelligent microplastic analysis system that analyzes fluorescence microscopy images to detect microplastics, estimate particle concentration, assess contamination risk, and generate downloadable analytical reports.
- Engineered a modular computer vision pipeline integrating image quality assessment, preprocessing, fine-tuned YOLOv8 inference, feature extraction, HSV-geometry validation, concentration estimation, and contamination risk scoring for reliable real-world image analysis.
- Designed a hybrid HSV-geometry validation layer that separated detection from validation, reducing false positives while preserving recall through domain-specific color and geometric filtering.
- Built an interactive React dashboard supporting single and batch image analysis, real-time result visualization, PDF report generation, and contextual guidance for interpreting contamination metrics.

## HACKATHONS

Udhgam Tensor v2.0 (24-Hour National Hackathon)

- Developed SmartMetric and advanced to the final round.

## CONTACT

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📄 [PORTFOLIO](#)

## EDUCATION

**B.Tech in Computer Science Engineering (AI & ML)**

**Woxsen University**

Expected Graduation: 2029

## TECHNICAL SKILLS

### Languages

- TypeScript, JavaScript, Python, SQL

### Frontend

- React, Next.js, Vite, Tailwind CSS, CSS Modules

### Backend

- FastAPI, Next.js Server Actions, Prisma ORM, NextAuth.js (Auth.js), Zod

### Database

- PostgreSQL

### ML & Computer Vision

- YOLOv8, OpenCV, PyTorch, NumPy

### Tools

- Git, GitHub, Bun

